

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Investigated and resolved complex hardware faults in **rack-level GPU platforms** that automation could not diagnose, correlating system and kernel logs with **BMC Redfish API** data.
- Automated routine fleet management and maintenance operations by developing **Python** scripts for hardware diagnostics and provisioning, reducing manual intervention by approximately **40%**.
- Collaborated with **Data Centre Operations** and **Hardware Engineering** teams to repair and remediate failed hardware, ensuring consistent delivery of new compute resources to customers.
- Participated in an on-call rotation, providing follow-the-sun coverage and partnering with US teams to address critical infrastructure issues in real-time.
- Built and documented processes for validating and testing new, **cutting-edge server hardware** and GPUs before they were fully vetted, ensuring peak efficiency for customer workloads.
- Developed **Linux shell scripts** and utilized **cron jobs** for system monitoring and log analysis, contributing to the stability of the AI infrastructure.
- Engineered **Python** scripts to streamline data pipelines and support scheduled reporting, integrating with **AWS S3** for data storage.

SKILLS

Infrastructure – Server Hardware, GPU Hardware, Data Center Operations, Hardware Engineering, Capacity Planning, Fleet Operations, Hardware Diagnostics, Hardware Repair, Server Provisioning, Linux Internals, Terminal Usage, Networking Fundamentals, Firmware Bugs, Error Message Analysis

Programming Languages – Python, Go, Shell Scripting

Cloud & Virtualization – AWS (EC2, S3, Lambda, CloudWatch), Docker

Operating Systems – Linux

APIs & Protocols – BMC Redfish API

Databases – PostgreSQL, MySQL, Oracle

Development Tools – Git, Jenkins, Jira, Confluence

PROJECTS

GPU Fleet Automation and Diagnostics Tooling

- Developed a **Python**-based tool to automate the diagnosis of common hardware faults across a fleet of **GPU servers**, reducing average troubleshooting time by **35%**.
- Integrated the tool with **BMC Redfish APIs** to gather low-level hardware status, enabling remote identification of component failures such as RAM or power supply issues.
- Created comprehensive documentation and runbooks for the new tooling, facilitating rapid adoption by the **Data Center Operations** team.

Linux System Monitoring and Log Analysis Automation

- Engineered a suite of **Linux shell scripts** and **Python** scripts to aggregate and analyze system logs from thousands of nodes, identifying recurring error patterns.
- Configured automated alerts using **AWS CloudWatch** for critical system events, improving incident detection latency by approximately **50%**.
- Implemented scheduled reporting via **Python** scripts to summarize system health and resource utilization for operations management.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)